

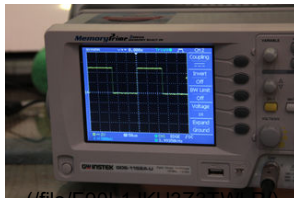
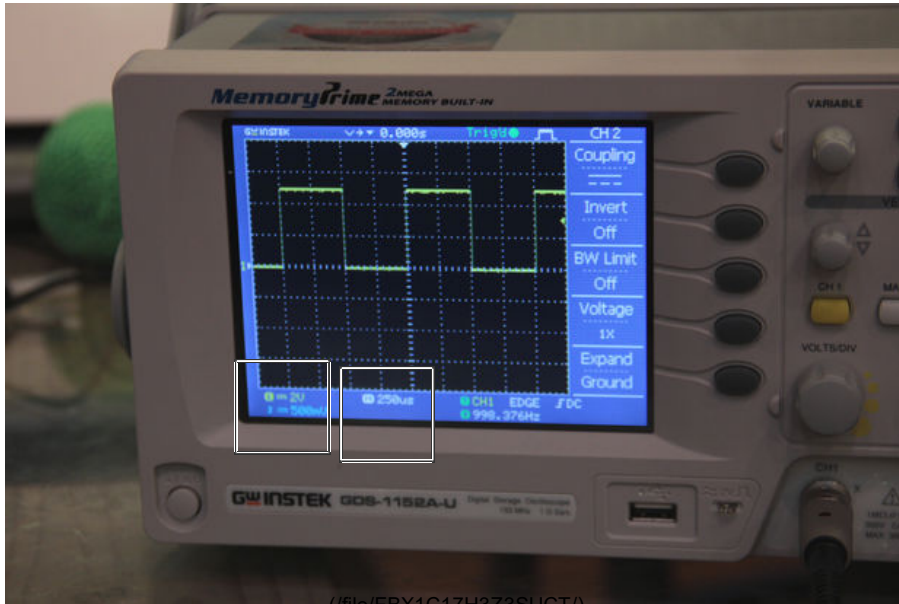
Arduino Timer Interrupts by

amandaghassaei (/member/amandaghassaei/) in [arduino \(/tag/type-id/category-technology/channel-arduino/\)](/tag/type-id/category-technology/channel-arduino/)[Download](#) [\(/id/Arduino-Timer-Interrupts/?ALLSTEPS\)](/id/Arduino-Timer-Interrupts/?ALLSTEPS)

6 Steps

 [\(/id/Arduino-Timer-Interrupts/step3/Example-1-Bike-Speedometer/\)](/id/Arduino-Timer-Interrupts/step3/Example-1-Bike-Speedometer/)[+ Collection](#)[I Made it!](#)[♥ Favorite](#)[🔊 Share ▾](#)

Step 2: Structuring Timer Interrupts

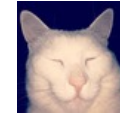


CS00	CS01	CS00	Description
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk_0 (No prescaler)
0	1	0	clk_0 (From prescaler)
0	1	1	clk_0 (From prescaler)
1	0	0	clk_0 (From prescaler)
1	0	1	clk_0 (From prescaler)
1	1	0	External clock source on T0 pin. Clock on falling edge.
1	1	1	External clock source on T0 pin. Clock on rising edge.

About This Instructable

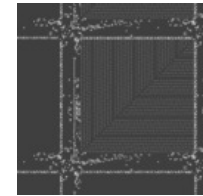
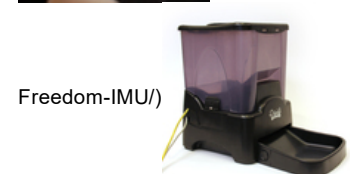
[👁 779.990 views](#)[♥ 537 favorites](#)

License:

[\(cc\) BY-NC-SA](#)**amandaghassaei**
(/member/amandaghassaei/)
uh-man-duh-guss-eye-dot-com
(http://www.amandaghassaei.com) [\(/member/amandaghassaei/\)](/member/amandaghassaei/)

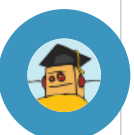
Bio: I'm a grad student at the Center for Bits and Atoms at MIT Media Lab. Before that I worked at Instructables, writing code for ... [More »](#) (/member/amandaghassaei/)

More by amandaghassaei:

 [\(/id/OTCA-](/id/OTCA-)[Metapixel-Conways-Game-of-Life/\)](#) [\(/id/9-Degrees-of-](/id/9-Degrees-of-)[Freedom-IMU/\)](#)[Timer setup code is done inside the `setup\(\)` function in an Arduino sketch.](/id/Twitter-Controlled-Pet-Feeder/)</div><div data-bbox=)

The code involved for setting up timer interrupts is a little daunting to look at, but it's actually not that hard. I pretty much just copy the same main chunk of code and change the prescaler and compare match register to set the correct interrupt frequency.

The main structure of the interrupt setup looks like this:



```
//https://www.instructables.com/id/Arduino-Timer-Interrupts/

void setup(){

cli();//stop interrupts

//set timer0 interrupt at 2kHz
TCCR0A = 0;// set entire TCCR0A register to 0
TCCR0B = 0;// same for TCCR0B
TCNT0 = 0;//initialize counter value to 0
// set compare match register for 2khz increments
OCR0A = 124;// = (16*10^6) / (2000*64) - 1 (must be <256)
// turn on CTC mode
TCCR0A |= (1 << WGM01);
// Set CS01 and CS00 bits for 64 prescaler
TCCR0B |= (1 << CS01) | (1 << CS00);
// enable timer compare interrupt
TIMSK0 |= (1 << OCIE0A);

//set timer1 interrupt at 1Hz
TCCR1A = 0;// set entire TCCR1A register to 0
TCCR1B = 0;// same for TCCR1B
TCNT1 = 0;//initialize counter value to 0
// set compare match register for 1hz increments
OCR1A = 15624;// = (16*10^6) / (1*1024) - 1 (must be <65536)
// turn on CTC mode
```

Notice how the value of OCR#A (the compare match value) changes for each of these timer setups. As explained in the last step, this was calculated according to the following equation:

compare match register = [16,000,000Hz/ (prescaler * desired interrupt frequency)] - 1

remember that when you use timers 0 and 2 this number must be less than 256, and less than 65536 for timer1

Also notice how the setups between the three timers differ slightly in the line which turns on CTC mode:

TCCR0A |= (1 << WGM01); //for timer0

TCCR1B |= (1 << WGM12); //for timer1

TCCR2A |= (1 << WGM21); //for timer2

This follows directly from the datasheet of the ATMELE 328/168.

Finally, notice how the setup for the prescalers follows the tables in the last step (the table for timer 0 is repeated above),

TCCR2B |= (1 << CS22); // Set CS#2 bit for 64 prescaler for timer 2

TCCR1B |= (1 << CS11); // Set CS#1 bit for 8 prescaler for timer 1

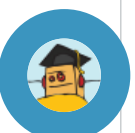
TCCR0B |= (1 << CS02) | (1 << CS00); // Set CS#2 and CS#0 bits for 1024 prescaler for timer 0

Notice in the last step that there are different prescaling options for the different timers. For example, timer2 does not have the option of 1024 prescaler.

The commands you want to execute during these timer interrupts are located in the Arduino sketch encapsulated in the following:

```
ISR(TIMER0_COMPA_vect){ //change the 0 to 1 for timer1 and 2 for timer2
//interrupt commands here
}
```

This bit of code should be located outside the setup() and loop() functions. Also, try to keep the interrupt routine as short as possible, especially if you are interrupting at a high frequency. It may even be worth addressing the ports/pins of the ATMELE chip directly instead of using the digitalWrite() and digitalRead() functions. You can find more info about that here (<http://www.arduino.cc/en/Reference/PortManipulation>).



Example- the following sketch sets up and executes 3 timer interrupts:

```
//timer interrupts
//by Amanda Ghassaei
//June 2012
//https://www.instructables.com/id/Arduino-Timer-Interrupts/

/*
 * This program is free software; you can redistribute it and/or modify
 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation; either version 3 of the License, or
 * (at your option) any later version.
 *
 */

//timer setup for timer0, timer1, and timer2.
//For arduino uno or any board with ATME1 328/168.. diecimila, duemilanove, lil
ypad, nano, mini...

//this code will enable all three arduino timer interrupts.
//timer0 will interrupt at 2kHz
//timer1 will interrupt at 1Hz
//timer2 will interrupt at 8kHz

//storage variables
boolean toggle0 = 0;
boolean toggle1 = 0;
boolean toggle2 = 0;
```

The images above show the outputs from these timer interrupts. Fig 1 shows a square wave oscillating between 0 and 5V at 1kHz (timer0 interrupt), fig 2 shows the LED attached to pin 13 turning on for one second then turning off for one second (timer1 interrupt), fig 3 shows a pulse wave oscillating between 0 and 5V at a frequency of 4khz (timer2 interrupt).

« Previous (/id/Arduino-Timer-Interrupts/)

Next » (/id/Arduino-Timer-Interrupts/step3/Example-1-Bike-Speedometer/)

Download

View All Steps (/id/Arduino-Timer-Interrupts/?ALLSTEPS)



We have a be nice comment policy.
Please be positive and constructive.

I Made it!

Add Images

Post Comment



AravindP4 (/member/AravindP4)

8 days ago

Reply

thank u sir



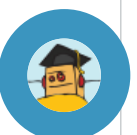
andreondra (/member/andreondra)

5 months ago

Reply

Great instructable! Thank you very much!

MalachiB1 (/member/MalachiB1) Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)





Thank you for breaking this down. I've read a lot of articles about timers but only yours makes it clear!

a year ago

[Reply](#)



dineshc31 (/member/dineshc31) ▶ [MalachiB1 \(/member/MalachiB1\)](#)

[Reply](#)

how exactly this is working for you all ?

6 months ago

For me while compiling in arduino 1.6.10 it is showing

error: 'TIMSK1' was not declared in this scope which is same for all above registers. do we need any `#include` header file here?



KennyL15 (/member/KennyL15)

9 months ago

[Reply](#)

Should also mention that the clock prescaler bit names CSX0, CSX1, CSX2 has the first number as 'X'. That number 'X' is linked to the timer number. So if 'X' is 0, then CS00, CS01, CS02 are prescaler bit names for timer 0; and the second value represents the bit position. So C00 means bit 0 (right-most-bit) of the 8-bit register TCCR0B, which is for timer 0.

The reason why I'm indicating this is because the tables (diagram) that go with "prescalers and compare match register" is confusing - since there's no mention about what CS10, CS11, CS12 etc are actually related to.



KennyL15 (/member/KennyL15)

9 months ago

[Reply](#)

"so a counter will take 10/16,000,000 seconds to reach a value of 9 (counters are 0 indexed)"

Do you actually mean it takes 10/16000000 seconds to begin (initial state) from index 0 and get back to a 0 index again? That is, initial index of 0. Then first increment is 1, second increment is 2, third increment is 3, all the way to 9th increment to index 9, and then one extra increment to get back to index 0 again. Altogether 10 increments to 'get back to 0'.



PeterMortensen (/member/PeterMortensen)

11 months ago

[Reply](#)

The comment for the line containing `ISR(TIMER2_COMPA_vect)` is confusing. **"timer1"** should probably be **"timer2"**.



Sharmila07 (/member/Sharmila07)

a year ago

[Reply](#)

my signal is in the frequency range between 20Hz and 500Hz, so i am taking the sampling frequency as 5000Hz (sampling time will be .2ms), after sampling i need to store the sampled values in the SD card and so i need to use spi. i want to know how much time will spi take to store in SD card and will i be able to sample my signal at this rate? kindly help



ParthJ5 (/member/ParthJ5)

a year ago

[Reply](#)

Can anyone please explain the connections to arduino in example 1 ?
I want to know how is the signal generator and Oscilloscope is connected to the Arduino ?

Thanks



★ **Bruce McCarthy (/member/Bruce McCarthy)**

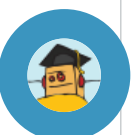
a year ago

[Reply](#)

Brilliant! I struggled with the datasheets for a few days before finding this instructable, and now it's all clear.

Thanks for sharing.

✕ [Check out our new classes! >> \(/classes/?utm_medium=cta&utm_source=banner\)](#)





avionics2 (/member/avionics2)

a year ago

Reply

This is the best interrupts explanations I have come across.

Thanks !



jerrychee94 (/member/jerrychee94)

a year ago

Reply

Hi, currently im doing project where it prompt user in serial communication and doing interrupt ISR at the same time. So , when user enter a specific value , it will stop the interrupt for few seconds and then restart the interrupt again. However , when restart the interrupt , the timing is out.(expected interrupt time : 5 microsec per interrupt , actual interrupt time : higher than 5 microsec)

my code is here..

```
void setup()
```

```
{
```

```
cli();
```

```
TCCR0A = 0;
```

```
TCCR0B = 0;
```

```
TCNT0 = 0;
```

```
OCR0A = 80;// = (16*10^6) / (200000*1) - 1 (must be <256)
```

```
//200000hz=5microsec
```

```
TCCR0A |= (1 << WGM01);
```

```
TCCR0B |= (0 << CS02)|(0 << CS01) | (1 << CS00);
```

```
TIMSK0 |= (1 << OCIE0A);
```

```
sei();
```

```
}
```

```
void loop()
```

```
{
```

```
if(serial.available>0)
```

```
{
```

```
//if user press something...
```

```
TIMSK0 &= ~(1 << OCIE0A);
```

```
delay(10000);
```

```
TIMSK0 |= (1 << OCIE0A);
```

```
}
```

```
}
```

```
ISR(TIMER0_COMPA_vect)
```

```
{
```

```
//code
```

```
}
```



PiraC (/member/PiraC)

a year ago

Reply

Good Work. Thank you!



sam.gwilliam (/member/sam.gwilliam)

a year ago

Reply

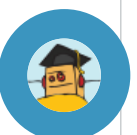
```
// turn on CTC mode
```

```
TCCR0A |= (1 << WGM01);
```

```
// Set CS01 and CS00 bits for 64 prescaler
```

```
TCCR0B |= (1 << CS01) | (1 << CS00);
```

Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)



What is the TCCR0A register? TCCR0B must be the 3-bit one as illustrated (where you set the prescaler) but I don't recall any mention of the other one.



MichealM1 (/member/MichealM1)

a year ago

Reply

This made it much clearer than other reads I have been to. Based on this explanation, I was able to devise a routine to discipline my software clock based on the 16MHz crystal. One of my units (Uno) is ticking at a very wrong speed. Gaining about 30 seconds per day. That's a LOT! But the software can now take care of it and any other errant crystal it encounters. Thanks!!



MichealM1 (/member/MichealM1) ▶ MichealM1 (/member/MichealM1)

Reply

a year ago

Sorry. Left out that I am getting NTP time hacks once per day but losing 30 seconds per day is way overboard. Maybe I will replace the crystal and see but as long as I can correct for it, it's not so bad.



Baguda (/member/Baguda) made it!

a year ago

Reply

In the code you set the timer1 register to CTC mode by "TCCR1B |= (1 << WGM12);". However, for all of the examples using timer0 and timer2, you set the CTC mode by "TCCR0B |= (1 << WGM01);". why do you set bit1 high instead of bit2 for timers 0 and 2 to put them into CTC mode? I found this table online that lists the mode settings(see attached), and it states that (100) is CTC mode. However, it is only for timer1... Is it different for timers 0 and 2?

Timer/Counter	Mode	WGM	Mode	WGM	Mode	WGM
0	Normal	000	0	Normal	000	0
0	Phase Correct PWM	001	0	Phase Correct PWM	001	0
0	Fast PWM	010	0	Fast PWM	010	0
0	Fast PWM	011	0	Fast PWM	011	0
0	Fast PWM	100	0	Fast PWM	100	0
0	Fast PWM	101	0	Fast PWM	101	0
0	Fast PWM	110	0	Fast PWM	110	0
0	Fast PWM	111	0	Fast PWM	111	0
0	CTC	100	1	CTC	100	1
0	CTC	101	1	CTC	101	1
0	CTC	110	1	CTC	110	1
0	CTC	111	1	CTC	111	1
1	Normal	000	1	Normal	000	1
1	Phase Correct PWM	001	1	Phase Correct PWM	001	1
1	Fast PWM	010	1	Fast PWM	010	1
1	Fast PWM	011	1	Fast PWM	011	1
1	Fast PWM	100	1	Fast PWM	100	1
1	Fast PWM	101	1	Fast PWM	101	1
1	Fast PWM	110	1	Fast PWM	110	1
1	Fast PWM	111	1	Fast PWM	111	1
1	CTC	100	2	CTC	100	2
1	CTC	101	2	CTC	101	2
1	CTC	110	2	CTC	110	2
1	CTC	111	2	CTC	111	2
2	Normal	000	2	Normal	000	2
2	Phase Correct PWM	001	2	Phase Correct PWM	001	2
2	Fast PWM	010	2	Fast PWM	010	2
2	Fast PWM	011	2	Fast PWM	011	2
2	Fast PWM	100	2	Fast PWM	100	2
2	Fast PWM	101	2	Fast PWM	101	2
2	Fast PWM	110	2	Fast PWM	110	2
2	Fast PWM	111	2	Fast PWM	111	2
2	CTC	100	3	CTC	100	3
2	CTC	101	3	CTC	101	3
2	CTC	110	3	CTC	110	3
2	CTC	111	3	CTC	111	3

(<https://cdn.instructables.com/FK7/FFSP/IJIUVOFI/FK7FFSPIJIUVOFI.LARGE.jpg>)



AnthonyP7 (/member/AnthonyP7)

2 years ago

Reply

If the Arduino is executing some code within an interrupt does the counter continue to count up so that the next interrupt is correctly timed?



sandro2204 (/member/sandro2204) ▶ AnthonyP7 (/member/AnthonyP7)

Reply

a year ago

Yes it does. But you should keep your interrupt service routine (ISR) as short as possible. Store a variable, set a flag and that's it. Do time consuming stuff in your main loop.



hal4512 (/member/hal4512)

a year ago

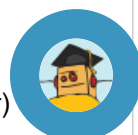
Reply

according to datasheet of atmega328p p106 -108 when we choose values of COM0A0 and COM0B0 for timer0

we set the behavior of pins OC0A OCOB After having config them as output and hardware will proceed as its describe in the datasheet.

all modes Normal mode ,CTC mode non PWM ..ect ..are described in page 100

when you'll read:For generating a waveform output in CTC mode the OC0A output can be set to toggle each logical level on each compare match (you don't have to toggle it by soft).





hal4512 (/member/hal4512)

a year ago

[Reply](#)

you don't need to write a code for toggling because atmega328p has 3 timers and every timer has 2 output pins ex: timer2 has OC2A pin 11 and OC2B pin3 and in isr routine u can do other thing.



KevinK91 (/member/KevinK91) ▶ hal4512 (/member/hal4512)

a year ago

[Reply](#)

could you please elaborate more on why the toggle code isn't necessary in this sketch?



hackerh (/member/hackerh)

a year ago

[Reply](#)

so if you wanted an interrupt every second (frequency of 1Hz):

compare match register = $[16,000,000 / (\text{prescaler} * 1)] - 1$
with a prescaler of 1024 you get:



metrog (/member/metrog)

a year ago

[Reply](#)

Hi,

Your project and explanations helped me a lot starting with my project. Thanks!

I'm using Arduino uno timer 0 and 2 to output 2 distinct (in sync) MIDI Clock signal and it works like charm. I can change BPM with a knob going from 30 to 250.

I'm struggling on my next step which is introducing an offset (-200ms / +200ms) between the 2 signals using an analogRead on a knob.

Would you know how to induce such an offset for a Serial output?

Timr0 and Timer2 doesnt have much room to change start point and end point and the induced delay would only be a few milliseconds maybe.

Well if this message finds you and you wish to reply, I'm all ears opened.

Thanks again,

Stef



tejonbiker (/member/tejonbiker)

a year ago

[Reply](#)

Wonderful, I worked with PICs but I see the AVR work slightly different



youraagh (/member/youraagh)

2 years ago

[Reply](#)

Hi, great tutorial, thanks heaps.

What libraries do I need to include, I'm getting compiler errors saying it doesn't know the names of the timing registers.



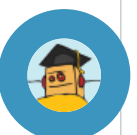
PravinahS (/member/PravinahS)

2 years ago

[Reply](#)

Hi Amanda,

I found these article very informative. I'm designing a card reader and i have included these timer interrupts in my programs.I have several confusions to use these timer interrupts on Arduino Leonardo.I have attached pins on pin 3 and pin 2 for D0 and D1 from my Mifare card reader.My card is a Mifare (32bits) type, so when i batch these card,i'm getting the extra bit of 33,so i decided to use timer interrupts just to execute exact 32 bits,but still i'm getting these extra bit.Please help me..Im lost!!





SitthinutK (/member/SitthinutK)

2 years ago

[Reply](#)

Thanks!!



RosmiR (/member/RosmiR)

2 years ago

[Reply](#)

In an Arduino code for pulse sensor 256 prescaler is activated by TCCR2b=0x06 . Why is it so ? Why is it not 0x04 as per the table for Timer0. Is it possible to make an interrupt of 100 Hz using Timer 2. (ocr2a <256, 1024 prescaler is not present). Could you help me with this. I am just a beginner.



RosmiR (/member/RosmiR)

2 years ago

[Reply](#)

How could I make an interrupt of 100Hz with timer 2. Could you share the table for timer2 similar to the one for Timer0 .



ProfRobNewton (/member/ProfRobNewton)

2 years ago

[Reply](#)

Just wondering if it would not be easier and more accurate to generate an interrupt when a switch change is registered and use millis() to determine the time since the last switch change? Sampling every 1ms using the method above may miss a switch change??



andycomic (/member/andycomic) ▶ **ProfRobNewton (/member/ProfRobNewton)**

2 years ago

[Reply](#)

I am no expert but to my knowledge the problem with switch state interrupts is that they bounce and your can't denounce a switch using software on an interrupt routine as its already been triggered (as in the interrupt routine is already done before you have a chance to de-bounce.). The other way is to use a hardware based button debounce switch which just uses a capacitor. This will solve the problem. However unless your project is huge, polling a switch every x cycles doesn't really make any difference to any project.



Fardenco (/member/Fardenco) ▶ **andycomic (/member/andycomic)**

2 years ago

[Reply](#)

I had the problem with a motor encoder, just put a 10k Ohms resistor to Vcc and a 1µF capacitor to GND around your interrupt pin, this will be ok !!
Pin interrupt would be more efficient because you'll never miss a change, and above that, you'll not waste time with useless interruption =)



DavidS13 (/member/DavidS13) ▶ **ProfRobNewton (/member/ProfRobNewton)**

2 years ago

[Reply](#)

While I don't think sampling every 1ms may miss a switch change (think about it, 1000 times a second???) I too would like to know if the more efficient way would be to make an interrupt for the switch change. I have a project that would use something similar and I'm trying to determine the most efficient course of action.



manoch_dc (/member/manoch_dc)

2 years ago

[Reply](#)

thank for Example interrupts that working.

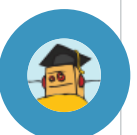
!!But void loop() has not work

my increase program

void setup(){

//set pins as outputs

✕ [Check out our new classes! >> \(/classes/?utm_medium=cta&utm_source=banner\)](#)




```

pinMode(7, OUTPUT); //my led
pinMode(8, OUTPUT);
pinMode(9, OUTPUT);
pinMode(13, OUTPUT);
cli();//stop interrupts
same example
sei();//allow interrupts
}

void loop(){
digitalWrite(7, HIGH); // turn the LED on (HIGH is the voltage level)
delay(100); // wait for a second
digitalWrite(7, LOW); // turn the LED off by making the voltage LOW
delay(100); // wait for a second
//do other things here
}

```



coltonschlosser (/member/coltonschlosser)

2 years ago

Reply

I made a python script that walks you through setting up the code for timer interrupts.
<https://gist.github.com/cltnschlosser/e747efae07a76e117e02>



killercatfish (/member/killercatfish)

2 years ago

Reply

Could this be used for a 2 player reflex timer game? Trying to push your players button on multiples of 5 seconds and getting a point when you do.



zinedine (/member/zinedine)

2 years ago

Reply

Good job, I want to import 200khz frequency, it's about 5us, but I only have 13.6us with every OCRnA that I used to change from 0 to 80. I'm using 2560. How can I do that?

Thanks



faizal91 (/member/faizal91)

2 years ago

Reply

Thank you... Very nice tutorial!



EnekoL (/member/EnekoL)

2 years ago

Reply

Hei, very usefull application, but i've one cuestion, how can i call de interrup for timer 1 inside the loop??Thanks!



ohoilett (/member/ohoilett) ▶ EnekoL (/member/EnekoL)

2 years ago

Reply

I could be wrong. You may want to check

<http://arduino.cc/en/Reference/AttachInterrupt>

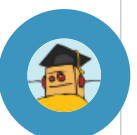


ohoilett (/member/ohoilett) ▶ EnekoL (/member/EnekoL)

2 years ago

Reply

You don't call interrupts like normal functions. An interrupt service routine (ISR) is "called" or better yet "triggered" at the specified frequency that you determined when you set up the interrupt.



★  **thegoodhen (/member/thegoodhen)** 3 years ago [Reply](#)

Hello... The code is very useful, however, be aware that the sketch you posted doesn't seem to ever get into the main loop!

 **DavidS13 (/member/DavidS13)** ▶ thegoodhen (/member/thegoodhen) 2 years ago [Reply](#)

I'm kind of a noob about all this, but unless I'm mistaken the point of interrupts is that you don't NEED a main loop. The main loop just runs forever and everything is done when interrupts happen. It is an extremely efficient way to code....

 **Unidentified Identity (/member/Unidentified Identity)** 3 years ago [Reply](#)

can we generate square wave with varying pulse width with this interrupt ?

★  **amandaghassaei (/member/amandaghassaei)** (author) ▶ Unidentified Identity (/member/Unidentified Identity) 3 years ago [Reply](#)

yes it's possible, but probably easier to just use the analogWrite command

 **srinivas_arduino (/member/srinivas_arduino)** 3 years ago [Reply](#)

sir please send program for generating 15000 sample values in 10 sec using timer interrupt values adc values from accelerometer sensor

 **srinivas_arduino (/member/srinivas_arduino)** 3 years ago [Reply](#)

sir please send program for generating 15000 sample values using timer interrupt values adc values from accelerometer sensor

 **AbdulW87 (/member/AbdulW87)** 3 years ago [Reply](#)

i am confused about something in the code, the line

```
// turn on CTC mode
```

```
TCCR0A |= (1 << WGM01);
```

shouldnt it be `TCCR0A |= (1 << WGM02);` or am i understanding this wrong, i am a beginner at arduino so its still very confusing to me.

 **O-Zo (/member/O-Zo)** 3 years ago [Reply](#)

Thank you for this tutorial, I'm struggling with a project and this helped me stumble at least to the right direction.

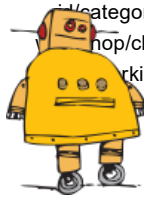
↓ More Comments

FEATURED CHANNELS

✕ Check out our new classes! >> (/classes/?utm_medium=cta&utm_source=banner)



[Woodworking \(/tag/type-woodworking/\)](#)[Paper \(/tag/type-paper/\)](#)[Kitchen Hacks \(/tag/type-kitchen%20hacks/\)](#)[Puzzles \(/tag/type-puzzles/\)](#)[Laser Cutting \(/tag/type-laser-cutting/\)](#)[Space \(/tag/type-space/\)](#)[Homesteading \(/tag/type-homesteading/\)](#)[3D Printing \(/tag/type-3d-printing/\)](#)[Sewing \(/tag/type-sewing/\)](#)



Newsletter

Join 2 million + to receive our paper!
DIY inspiration in your inbox.

About Us

[Who We Are \(/about/\)](#)[Advertise \(/advertise/\)](#)[Contact \(/about/contact.jsp\)](#)[Jobs \(/community/Positions-available-at-Instructables/\)](#)[Help \(/id/how-to-write-a-great-instructable/\)](#)

Find Us

[Facebook \(http://www.facebook.com/instructables\)](http://www.facebook.com/instructables)[Youtube \(http://www.youtube.com/user/instructablestv\)](http://www.youtube.com/user/instructablestv)[Twitter \(http://www.twitter.com/instructables\)](http://www.twitter.com/instructables)[Pinterest \(http://www.pinterest.com/instructables\)](http://www.pinterest.com/instructables)[Google+ \(https://plus.google.com/+instructables\)](https://plus.google.com/+instructables)

Resources

[For Teachers \(/teachers/\)](/teachers/)[Artists in Residence \(/air\)](/air)[Gift Premium Account \(/account/give?source=footer\)](/account/give?source=footer)[Forums \(/community/\)](/community/)[Answers \(/tag/type-question/?sort=RECENT\)](/tag/type-question/?sort=RECENT)[Sitemap \(/sitemap/\)](/sitemap/)